



CERTIFICATE OF TESTS

Heat: 24102171
Material: 5L- X52 4-1/2X.337 (15.00#)BR,DRL,PEB (11000102)
Vendor: Big River Steel - Osceola AR US (180431)

Purchase Order: 26708947
Sales Order: 1001001265
MFG Date: 09/24/2024

CHEMICAL PROPERTIES

	C	MN	P	S	NI	CR	CU	MO	SI	TI	V	CE
Ladle	0.0600	1.0400	0.0100	0.0010	0.0400	0.0500	0.0900	0.0170	0.1900	0.0010	0.0030	0.1284
	Al: 0.0390		Sn: 0.0046		N: 0.0077		B: 0.0002		Ca: 0.0033		Nb: 0.0020	
Chem	0.0633	1.0734	0.0089	0.0017	0.0364	0.0421	0.0644	0.0105	0.1806	0.0010	0.0045	0.1301
	Al: 0.0356		Sn: 0.0038		N: 0.0560		B: 0.0000		Ca: 0.0022		Nb: 0.0019	
Chem	0.0647	1.0820	0.0088	0.0017	0.0371	0.0411	0.0652	0.0108	0.1829	0.0009	0.0049	0.1324
	Al: 0.0375		Sn: 0.0040		N: 0.0164		B: 0.0001		Ca: 0.0020		Nb: 0.0027	

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 73.09 TS (ksi): 77.78 EL (%) : 26.00 HRB: 86.00 Y/T(%): 0.94

CHARPY TEST RESULTS (BODY)

Orientation : TRANSVERSE	Specimen(S)	Energy (ft-lb)	%Shear
Spec. Size (in) : .394"X.197"X2.165"	1	87.00	100.00
Test Temp (°F) : -20	2	90.00	100.00
Location : 3T90	3	85.00	100.00
	Average	87.33	100.00

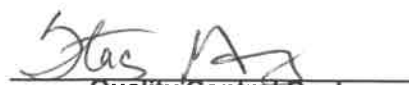
Acceptance Criteria Min. (ft-lb) : 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-24,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Big River Steel - Osceola AR US
COIL MANUFACTURER: Big River Steel - Osceola AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.


Quality Control Dept